# Yuyuan Han

Computational imaging & physics-aware learning

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PhD student in computational imaging and physics-aware learning from optical measurements.

## EDUCATION & RESEARCH APPOINTMENTS

**Xi'an Jiaotong University** · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

**Northwest University** · B.Eng. — 2018–2022
Electronic Information Science and Technology

**Politecnico di Milano** · Visiting PhD student — Oct 2026–present
Department of Physics · Host: Antonio Pifferi

## SELECTED RESEARCH

### The Lift Spectrum · Robust image-free perception

Preprint
arXiv · 2026 · Paper · Code

- Showed that segmenting directly from 3.13%-sampled single-pixel measurements beats the strongest clean-trained reconstruct-then-segment baseline by 3.1–21.7 foreground-mIoU points under 20 dB noise.
- Designed a content-adaptive measurement-to-space lift trained with scheduled auxiliary reconstruction, exceeding the prior image-free baseline by 3.2–9.9 points at the same sampling rate.

### Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Designed tunable-sparsity Hadamard illumination with two acquisition modes; derived a sparsity-noise model and validated noise resilience in simulations and an optical experiment through ground glass.

### LDGI · Optical phased-array imaging

Optics Communications · 2025 · Paper · Code

- Showed that line patterns from one-dimensional optical phased arrays break the noise symmetry assumed by traditional ghost-imaging denoising; developed a loop differential algorithm and validated it optically.

## CURRENT RESEARCH

Developing acquisition-aware models and adapting pretrained multimodal models to bucket-detector signals. Planned visiting research: time-domain diffuse optics and 3D reconstruction.

## TOOLS & OPEN SOURCE

**Tools** · Python, PyTorch, MATLAB, COMSOL; open-source remote-gpu-trainer for reproducible remote-GPU experiments (60+ GitHub stars)

**Experiments** · DMD/SLM illumination, single-pixel detection, optical alignment and system integration

**Device modeling** · Electro-optic phased-array geometry modeling and optimization for beam steering

## AWARD

**2024** · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code